

Commit e0c92f0

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 ActuallyNull committed on Aug 10

added standardisation and normalisation for features in samples dataset

 main

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 data_processing.py

 1 file changed +33 -11 lines changed

Q Search within code



data_processing.py   +33 -11  ...

 ...

@@ -8,11 +8,24 @@

88

records = []

99

labels = []

1010

physionet_label_map = {'N': "Normal", 'A': "AFib", 'O': "Other Arrythmia"}

11

+ duration = 10 #seconds

1112

1213

X = []

1314

y = []

1415

15

- def load_physionet_dataset(physionet_path, reference_file_path):

16

+ def standardise_dataset(X):

17

+ X_scaled = np.array([(x - np.mean(x)) / np.std(x) for x in X])

18

+ return X_scaled

19

+

20

+ from sklearn.preprocessing import MinMaxScaler

21

+

22

+ def normalise_dataset(X):

23

+ # Reshape X to 2D if needed (n_samples, n_features)

24

+ scaler = MinMaxScaler()

25

+ X_norm = scaler.fit_transform(X)

26

+ return X_norm

27

+

28

+ def load_physionet_dataset(physionet_path, reference_file_path, X, y):

1629

1730

reference_df = pd.read_csv(reference_file_path, header=None, names=['record', 'label'])

1831

 ...

@@ -33,20 +46,29 @@ def load_physionet_dataset(physionet_path, reference_file_path):

3346

3447

entering it in dataset

3548

for paths in records:

36

- record = wfdb.rdrecord(physionet_path+"\\")+paths)

49

+ record = wfdb.rdrecord(physionet_path + "\\\" + paths)

50

+ fs = record.fs

51

+ samples = fs * duration

3752

ecg_signal = record.p_signal[:, 0]

38

- X.append(ecg_signal)

53

+ trunc_ecg = ecg_signal[:samples]

54

+ # Pad if shorter than 'samples'

55

+ if len(trunc_ecg) < samples:

56

+ trunc_ecg = np.pad(trunc_ecg, (0, samples - len(trunc_ecg)), 'constant')

57

+ X.append(trunc_ecg)

3958

4059

for label in labels:

4160

y.append(label)

4261

43

- """ def load_mitbih_dataset(mitbih_path):

44

- mitbih_records = []

62

+ X = np.array(X, dtype=np.float32)

63

+ y = np.array(y)

64

+ return X, y

4565

46

- #loading data

47

- for file in os.listdir(mitbih_path):

48

- if file.endswith(".hea"):

49

- record_name = file[:-4]

50

- mitbih_records.append(record_name) """

66

+ def process_physionet_dataset(physionet_path, reference_file_path, X, y): #X and y should be empty lists

67

+

68

+ X, y = load_physionet_dataset(physionet_path, reference_file_path, X, y)

69

+ X = standardise_dataset(X)

70

+ X = normalise_dataset(X)

71

+ return X, y

72

+

73

+ X, y = process_physionet_dataset("training2017/training2017", "training2017/training2017/REFERENCE.csv", X, y)

5174

52

-



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